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sukesh273@LAPTOP-PI08CKLL:~$ mkdir -p ~/OS-Third-Program
sukesh273@LAPTOP-PI08CKLL:~$ cd ~/OS-Third-Program
sukesh273@LAPTOP-PI08CKLL:~/OS-Third-Program$ pwd
/home/sukesh273/OS-Third-Program
sukesh273@LAPTOP-PI08CKLL:~/OS-Third-Program$ code .
sukesh273@LAPTOP-PI08CKLL:~/OS-Third-Program$ |
```

program3.py X

program3.py

```
1  from collections import deque
2
3  processes = [
4      {"pid": "P1", "arrival": 0, "burst": 7, "priority": 2},
5      {"pid": "P2", "arrival": 2, "burst": 4, "priority": 1},
6      {"pid": "P3", "arrival": 4, "burst": 1, "priority": 3},
7      {"pid": "P4", "arrival": 5, "burst": 4, "priority": 2},
8  ]
9
10
11 def priority_scheduling(processes):
12     current_time = 0
13     completed = set()
14     result = []
15
16     while len(completed) < len(processes):
17         ready = [
18             p for p in processes
19             if p["arrival"] <= current_time
20             and p["pid"] not in completed
21         ]
22
23         if not ready:
24             next_time = min(
25                 p["arrival"]
26                 for p in processes
27                 if p["pid"] not in completed
28             )
29
```

```
        result.append(("IDLE", current_time, next_time))
        current_time = next_time
        continue

    p = min(
        ready,
        key=lambda x: (
            x["priority"],
            x["arrival"],
            x["pid"]
        )
    )

    start = current_time
    end = start + p["burst"]

    result.append((p["pid"], start, end))

    current_time = end
    completed.add(p["pid"])

return result
```

```
def round_robin(processes, quantum):
    if quantum <= 0:
        print("Quantum must be greater than 0")
        return []

    process_list = sorted(
        processes,
        key=lambda x: (x["arrival"], x["pid"])
    )

    remaining = {
        p["pid"]: p["burst"]
        for p in process_list
    }

    queue = deque()
    result = []

    current_time = 0
    i = 0

    while i < len(process_list) or queue:
        if not queue:
            if current_time < process_list[i]["arrival"]:
                result.append((
```

```
        result.append((
            "IDLE",
            current_time,
            process_list[i]["arrival"]
        ))

        current_time = process_list[i]["arrival"]

    while (
        i < len(process_list)
        and process_list[i]["arrival"] <= current_time
    ):
        queue.append(process_list[i])
        i += 1

    p = queue.popleft()

    run_time = min(
        quantum,
        remaining[p["pid"]]
    )

    start = current_time
    current_time += run_time
```

```

        result.append((
            p["pid"],
            start,
            current_time
        ))

        remaining[p["pid"]] -= run_time

        while (
            i < len(process_list)
            and process_list[i]["arrival"] <= current_time
        ):
            queue.append(process_list[i])
            i += 1

        if remaining[p["pid"]] > 0:
            queue.append(p)

    return result

def show_result(title, result):
    print("\n" + title)
    print("Process Start End")

    for item in result:

```

```

9     for item in result:
10         print(item[0], item[1], item[2])
11
12     sequence = [item[0] for item in result]
13
14     print("Sequence:", " -> ".join(sequence))
15
16 print("INPUT DATA")
17 print("PID AT BT Priority")
18
19 for p in processes:
20     print(
21         p["pid"],
22         p["arrival"],
23         p["burst"],
24         p["priority"]
25     )
26
27 print("\nPriority Rule: Smaller number = Higher priority")
28
29 priority_result = priority_scheduling(processes)
30
31 show_result(
32     "NON-PREEMPTIVE PRIORITY",
33     priority_result
34 )

```

```
quantum = 2

print("\nRound Robin Quantum =", quantum)

rr_result = round_robin(
    processes,
    quantum
)

show_result(
    "ROUND ROBIN",
    rr_result
)
```

```
● suresh273@LAPTOP-PI08CKLL:~/OS-Third-Program$ pwd
/home/suresh273/OS-Third-Program
● suresh273@LAPTOP-PI08CKLL:~/OS-Third-Program$ python3 program3.py
INPUT DATA
PID AT BT Priority
P1 0 7 2
P2 2 4 1
P3 4 1 3
P4 5 4 2

Priority Rule: Smaller number = Higher priority

NON-PREEMPTIVE PRIORITY
Process Start End
P1 0 7
P2 7 11
P4 11 15
P3 15 16
Sequence: P1 -> P2 -> P4 -> P3

Round Robin Quantum = 2

ROUND ROBIN
Process Start End
P1 0 2
P2 2 4
P1 4 6
P3 6 7
P2 7 9
P4 9 11
P1 11 13
P4 13 15
P1 15 16
Sequence: P1 -> P2 -> P1 -> P3 -> P2 -> P4 -> P1 -> P4 -> P1
○ suresh273@LAPTOP-PI08CKLL:~/OS-Third-Program$
```

